An exponential-hierarchy oracle prevents almost-everywhere subexponential pseudodeterministic constructions

Shuichi Hirahara, Zhenjian Lu, Hanlin Ren

doi:10.4230/LIPIcs.CCC.2023.6; CC BY 4.0

Editorial note. Complete author official proof faithfully transcribed from CC BY4 published PDF. Author mathematics and original labels retained. Standard cited Find1 and probability-class facts are retained without recursive reproof. No native author TeX was located through the official/ECCC/author publication routes.

Allowed standard prerequisites (editorial).

- Pseudodeterministic Find1 query lower bound and BPP containment in the second level. Exact original Theorems51 [Goldwasser-Impagliazzo-Pitassi-Santhanam2021] and23 [Sipser1983] retained as author-cited standard inputs

Definitions and notations (original Sections 1.1, 2.1). For a complexity class $\mathfrak{C}$, a complexity-theoretic statement is $\mathfrak{C}$-relativizing if it is true relative to every oracle $\mathcal{O} \in \mathfrak{C}$.

Definition 20. A language L is in EXPH if there is a constant k, an exponential bound $e(n) = 2^{\mathrm{poly}(n)}$, and a $\mathrm{poly}(e(n))$-time algorithm M such that, for every input $x \in \{0,1\}^*$,

$$x \in L \iff \exists y_1 \forall y_2 \exists y_3 \forall y_4 \dots \exists y_k,\ M(x, y_1, \dots, y_k) = 1,$$

where $y_i \in \{0,1\}^{e(n)}$ for each $i \in [k]$. Equivalently, a language is in EXPH if it can be decided in exponential ($2^{\mathrm{poly}(n)}$) time with access to a PH oracle.

Pseudodeterministic algorithms. A pseudodeterministic algorithm is a probabilistic algorithm that returns a fixed output with high probability on any given input. We will also consider pseudodeterministic oracle (or advice-taking) algorithms. Such an algorithm will maintain its pseudodeterministic behavior (i.e., outputting a fixed answer with high probability for any input) when a correct oracle (or advice) is given.

Dense property (original Section 1.3.2). A property $Q \subseteq \{0,1\}^*$ is dense if for every input length $n \in \mathbb{N}$, we have that $|Q \cap \{0,1\}^n| \geq 2^n / \mathrm{poly}(n)$.

Theorem 23 [68]. For every oracle $\mathcal{O}$, $\mathsf{BPP}^{\mathcal{O}} \subseteq (\Sigma_2^p)^{\mathcal{O}}$.

5.2 Bounded-Relativization Barriers for Explicit Constructions

5.2.1 Barriers for Almost-Everywhere Pseudodeterministic Constructions

We show that EXPH-relativizing techniques cannot prove almost-everywhere pseudodeterministic constructions, even if the construction algorithms are allowed $2^{o(n)}$ time. The underlying oracle uses the query complexity lower bounds proved by Goldwasser, Impagliazzo, Pitassi, and Santhanam [31] in a black-box fashion. (We thank Rahul Santhanam for pointing out that the query complexity lower bounds imply an EXPH-computable oracle without almost-everywhere pseudodeterministic constructions in a black-box fashion.)

Definition 50. A pseudodeterministic decision tree for a search problem S is a distribution $\mathcal{T}$ over decision trees with the following property: For every input x, there is a canonical value o such that with probability at least $2/3$, $\mathcal{T}(x) = o$. Let $\mathrm{psPdt}(S)$ denote the minimum depth of any pseudodeterministic decision tree for S. Consider the following search problem denoted as Find1. The input is a string $x \in \{0,1\}^N$ where it is guaranteed that there are at least $N/2$ bits in x that are equal to 1. The problem is to find some $i \in [N]$ such that $x_i = 1$. The following lower bound on the pseudodeterministic query complexity of Find1 was proved in [31]:

Theorem 51 [31]. $\mathrm{psPdt}(\mathsf{Find1}) = \Omega(\sqrt{N})$. We use this result as a black box and construct an oracle $\mathcal{O} \in \mathsf{EXPH}$ without almost-everywhere pseudodeterministic constructions running in $2^{o(n)}$ time:

Theorem 52. There is an oracle $\mathcal{O} \in \mathsf{EXPH}$ and a dense property $Q \in \mathsf{P}^{\mathcal{O}}$ such that the following holds. For every randomized (and purportedly pseudodeterministic) algorithm A that runs in $2^{o(n)}$ time with oracle access to $\mathcal{O}$ and every infinite sequence of outputs $\{x_n\}_{n\in\mathbb{N}}$ where each $x_n \in Q \cap \{0,1\}^n$, there are infinitely many input lengths $n \in \mathbb{N}$ such that

$$\Pr[A(1^n) = x_n] \leq 3/4.$$

Proof. Let $M_1, M_2, \ldots$ be an enumeration of probabilistic machines running in $2^{0.1n}$ time. Let n_1 be a large enough constant and define the sequence $\{n_i\}$ where $n_i = 2^{n_{i-1}}$ for each $i \geq 2$. We let $Q = \mathcal{O}$ itself be the dense property without pseudodeterministic construction algorithms. In particular, we want that for every $i \in \mathbb{N}$, $M_i(1^{n_i})$ fails to generate a canonical string $x \in \mathcal{O} \cap \{0,1\}^{n_i}$. Naturally, our oracle construction proceeds in stages, where for each $i \in \mathbb{N}$, the n_i-th slice of $\mathcal{O}$ is constructed by carefully diagonalizing against M_i; on the other hand, if $n \in \mathbb{N}$ is not in the sequence $\{n_i\}$, then we simply let $\mathcal{O}$ accept every string of length n.

Now we show how to construct the n_i-th slice of $\mathcal{O}$ in the i-th stage. The idea is simple: construct the (purportedly pseudodeterministic) decision tree $\mathcal{T}$ corresponding to M_i, find an exponential-length input tt of Find1 on which $\mathcal{T}$ fails, and let the truth table of the n_i-th slice of $\mathcal{O}$ be tt. More precisely:

- **Converting M_i into a distribution of decision trees.** We define the distribution of decision trees $\mathcal{T}$. The input to $\mathsf{Find1}$ has length 2^{n_i} and is considered as the truth table of the n_i-th slice of $\mathcal{O}$. Note that since $2^{0.1n_i} \ll n_{i+1}$, for every $n' \leq 2^{0.1n_i}$ such that $n' \neq n_i$, the n'-th slice of $\mathcal{O}$ are fixed. To sample a (deterministic) decision tree T from $\mathcal{T}$, sample a sequence of random coins fed to M_i and simulate M_i on the oracle $\mathcal{O}$. Whenever M_i makes a query $\mathcal{O}(x)$, if $|x| = n_i$, then the decision tree asks the x-th bit of our input; otherwise we return the already-fixed value of $\mathcal{O}(x)$.
- **Invoking the lower bound.** Note that $\mathcal{T}$ only makes $2^{0.1n_i} < o(\sqrt{2^{n_i}})$ queries. By Theorem 51, there exists an input $tt \in \{0,1\}^{2^{n_i}}$ such that $\mathcal{T}$ fails to solve $\mathsf{Find1}$ on input tt pseudodeterministically. That is, for every valid output x such that $tt_x = 1$, $\Pr[\mathcal{T}(tt) = x] < 2/3$. Let the truth table of the n_i-th slice of $\mathcal{O}$ to be such a tt.

It is easy to see that for every probabilistic machine A that runs in $2^{o(n)}$ time with oracle access to $\mathcal{O}$, and every infinite sequence of outputs $\{x_n\}_{n\in\mathbb{N}}$ where each $x_n \in \mathcal{O} \cap \{0,1\}^n$, for every i such that $A = M_i$, the probability that $A(1^{n_i})$ outputs x_{n_i} is at most $3/4$.

Complexity of $\mathcal{O}$. Let $\mathcal{O}_{\leq n_{i-1}}$ denote the description of the oracle $\mathcal{O}$ up to the n_{i-1}-th slice (note that $\mathcal{O}_{\leq n_{i-1}}$ can be described in $\mathrm{poly}(2^{n_{i-1}}) \leq \mathrm{poly}(n_i)$ bits), and $x \in \{0,1\}^{n_i}$. Note that $\mathcal{O}_{\leq n_{i-1}}$ and tt together defines the oracle $\mathcal{O}$ up to input length $n_{i+1} - 1$, so the behavior of $M_i^{\mathcal{O}}$ is completely determined by M_i, $\mathcal{O}_{\leq n_{i-1}}$, and tt. Let $\mathsf{ProbEst}(\mathcal{O}_{\leq n_{i-1}}, tt, M_i, x)$ be an oracle such that

$$\Pr[M_i^{\mathcal{O}}(1^{n_i}) = x] > 3/4 \Longrightarrow \mathsf{ProbEst}(\mathcal{O}_{\leq n_{i-1}}, tt, M_i, x) = 1,$$
$$\Pr[M_i^{\mathcal{O}}(1^{n_i}) = x] < 2/3 \Longrightarrow \mathsf{ProbEst}(\mathcal{O}_{\leq n_{i-1}}, tt, M_i, x) = 0.$$

By Theorem 23, the oracle $\mathsf{ProbEst}$ can be implemented in $\mathsf{DTIME}[2^{O(n_i)}]^{\mathsf{PH}}$. Consider the following algorithm with oracle access to $\mathsf{ProbEst}$ that prints a truth table tt. The algorithm maintains the prefix of a truth table which is initially the empty string, and extends this prefix bit by bit. Suppose that we have a prefix of length ℓ, denoted as $tt' \in \{0,1\}^{\ell}$. To fix the $(\ell+1)$-st bit of tt', we check if there exists a truth table tt such that (1) for every $x \in \{0,1\}^{n_i}$, $\mathsf{ProbEst}(\mathcal{O}_{\leq n_{i-1}}, tt, M_i, x) = 0$, and (2) $tt' \circ 0$ (tt' concatenated with a bit 0) is a prefix of tt. If there is such a tt, then we append 0 to the end of tt'; otherwise we append 1 to the end of tt'.

It is clear that the algorithm always outputs a truth table tt on which M_i fails. Since $\mathsf{ProbEst}$ runs in $\mathsf{DTIME}[2^{O(n_i)}]^{\mathsf{PH}}$, the whole algorithm also runs in $\mathsf{DTIME}[2^{O(n_i)}]^{\mathsf{PH}}$. Using this algorithm, it is easy to construct the oracle $\mathcal{O}$ in EXPH. □

References

[31] Shafi Goldwasser, Russell Impagliazzo, Toniann Pitassi, and Rahul Santhanam. On the pseudo-deterministic query complexity of NP search problems. CCC2021, LIPIcs200,36:1–36:22. doi:10.4230/LIPIcs.CCC.2021.36.

[33] Shuichi Hirahara. Non-black-box worst-case to average-case reductions within NP. FOCS2018,247–258. doi:10.1109/FOCS.2018.00032.

[34] Shuichi Hirahara. Non-disjoint promise problems from meta-computational view of pseudorandom generator constructions. CCC2020,LIPIcs169,20:1–20:47. doi:10.4230/LIPIcs.CCC.2020.20.

[68] Michael Sipser. A complexity theoretic approach to randomness. STOC1983,330–335. doi:10.1145/800061.808762.